# DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: Helios Watch Store E-commerce Database

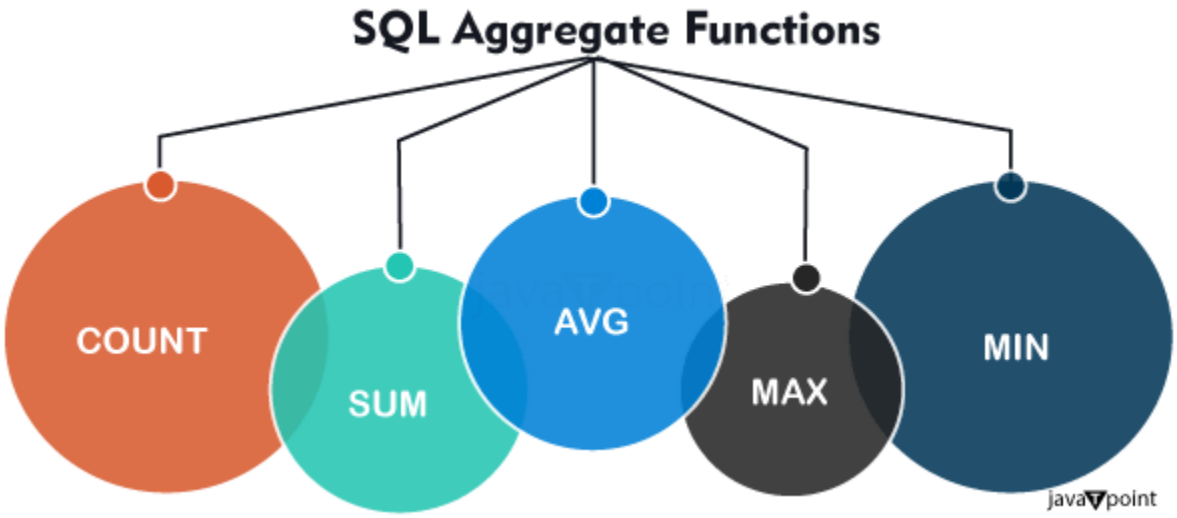
Skill Task: DBMS

Week: 9

## 1.Aggregate Functions

**Definition**

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.




### 1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM Orders;

TOTAL_ORDERS
------------
           9
```

### 1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      51891
```

### 1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(Total_Amount) AS Average_Order_Amount
  2  FROM Orders;

AVERAGE_ORDER_AMOUNT
--------------------
          5765.66667
```

### 1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(Total_Amount) AS Lowest_Order_Amount
  2  FROM Orders;

LOWEST_ORDER_AMOUNT
-------------------
               1999
```

### 1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(Total_Amount) AS Highest_Order_Amount
  2  FROM Orders;

HIGHEST_ORDER_AMOUNT
--------------------
               12999
```

## 2. GENERATE TOTAL SALES REPORTS

Display the total sales amount

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM Orders;

TOTAL_SALES
-----------
      51891
```

## 2.1 CUSTOMER-WISE SALES REPORT

```
SQL> SELECT c.Customer_ID,
  2  c.First_Name,
  3  c.Last_Name,
  4  SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name,
 10  c.Last_Name
 11  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- --------------------
          8 Swetha
Balaji                                                            12999

          5 Keerthana
Mohan                                                              7499

          1 Renu
Srinivasan                                                         6999


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- --------------------
          4 Rahul
Sharma                                                             5999

          7 Karthik
Prakash                                                            5299

          6 Nandhini
Ravi                                                               4599


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- --------------------
          2 Arjun
Kumar                                                              3999

          9 Vignesh
Ganesh                                                             2499

          3 Priya
Kumar                                                              1999


9 rows selected.
```

## 3. FIND TOP CUSTOMERS BASED ON PURCHASE AMOUNT

```
SQL> SELECT c.Customer_ID,
  2  c.First_Name,
  3  c.Last_Name,
  4  SUM(o.Total_Amount) AS Total_Purchase_Amount
  5  FROM Customer c
  6  INNER JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID,
  9  c.First_Name,
 10  c.Last_Name
 11  ORDER BY Total_Purchase_Amount DESC;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          8 Swetha
Balaji                                                          12999

          5 Keerthana
Mohan                                                            7499

          1 Renu
Srinivasan                                                       6999


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
```

```
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          4 Rahul
Sharma                                                           5999

          7 Karthik
Prakash                                                          5299

          6 Nandhini
Ravi                                                             4599


CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          TOTAL_PURCHASE_AMOUNT
-------------------------------------------------- ---------------------
          2 Arjun
Kumar                                                            3999

          9 Vignesh
Ganesh                                                           2499

          3 Priya
Kumar                                                            1999


9 rows selected.
```

## 4. IDENTIFY BEST-SELLING PRODUCTS

```
SQL> SELECT p.Product_ID,
  2  p.Product_Name,
  3  SUM(od.Quantity) AS Total_Quantity_Sold
  4  FROM Product p
  5  INNER JOIN Order_Details od
  6  ON p.Product_ID = od.Product_ID
  7  GROUP BY p.Product_ID,
  8  p.Product_Name
  9  ORDER BY Total_Quantity_Sold DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       103
Digital Sport Watch
                  4

       101
Classic Silver Watch
                  2

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
```

```
        105
Smart Active Watch
                    1

        106
Sports Chronograph

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
                    1

        107
Outdoor Pro Watch
                    1

        108

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
Royal Gold Edition
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
Royal Gold Edition
                  1

       102
Elegant Gold Watch
                  1


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       104
Smart Fit Pro
                  1


8 rows selected.
```

## 4.1 PRODUCT SALES AMOUNT

```
SQL> SELECT p.Product_ID,
  2  p.Product_Name,
  3  SUM(od.Quantity) AS Total_Quantity_Sold,
  4  SUM(od.Subtotal) AS Total_Sales_Amount
  5  FROM Product p
  6  INNER JOIN Order_Details od
  7  ON p.Product_ID = od.Product_ID
  8  GROUP BY p.Product_ID,
  9  p.Product_Name
 10  ORDER BY Total_Sales_Amount DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
       108
Royal Gold Edition
                  1              12999

       103
Digital Sport Watch
                  4               7996

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
```

```
       105
Smart Active Watch
                  1                7499

       104
Smart Fit Pro

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
                  1                5999

       107
Outdoor Pro Watch
                  1                5299

       101

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
Classic Silver Watch
                  2                4998
```

```
       106
Sports Chronograph
                  1                4599


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------
       102
Elegant Gold Watch
                  1                3999


8 rows selected.
```

## 5. CATEGORY-WISE SALES ANALYSIS

```
SQL> SELECT c.Category_ID,
  2  c.Category_Name,
  3  SUM(od.Quantity) AS Total_Quantity_Sold,
  4  SUM(od.Subtotal) AS Total_Sales_Amount
  5  FROM Category c
  6  INNER JOIN Product p
  7  ON c.Category_ID = p.Category_ID
  8  INNER JOIN Order_Details od
  9  ON p.Product_ID = od.Product_ID
 10  GROUP BY c.Category_ID,
 11  c.Category_Name
 12  ORDER BY Total_Sales_Amount DESC;
```

```
CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------

          3 Smart Watch
                  2              13498

          5 Luxury
                  1              12999

          4 Sports
                  2               9898


CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD TOTAL_SALES_AMOUNT
------------------- ------------------

          1 Analog
                  3               8997

          2 Digital
                  4               7996
```

## 6. COMBINED BUSINESS SALES REPORT

```
SQL> SELECT cu.Customer_ID,
  2  cu.First_Name,
  3  cu.Last_Name,
  4  o.Order_ID,
  5  o.Order_Date,
  6  p.Product_ID,
  7  p.Product_Name,
  8  cat.Category_Name,
  9  od.Quantity,
 10  od.Unit_Price,
 11  od.Subtotal
 12  FROM Customer cu
 13  INNER JOIN Orders o
 14  ON cu.Customer_ID = o.Customer_ID
 15  INNER JOIN Order_Details od
 16  ON o.Order_ID = od.Order_ID
 17  INNER JOIN Product p
 18  ON od.Product_ID = p.Product_ID
 19  INNER JOIN Category cat
 20  ON p.Category_ID = cat.Category_ID
 21  ORDER BY cu.Customer_ID,
 22  o.Order_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
```

```
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
         1 Renu

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Srinivasan                                             1001 15-SEP-26

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
```

```
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
       101

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Classic Silver Watch

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
```

```
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
         1 Renu

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Srinivasan                                             1001 15-SEP-26

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
```

```
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
         4 Rahul

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Sharma                                              1004 04-SEP-26

CUSTOMER_ID FIRST_NAME
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
       106

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Sports Chronograph

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
```

```
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                       QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
         9 Vignesh

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                           ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                       QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
Ganesh                                                  1009 09-SEP-26
```

```
CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------
      2499

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY UNIT_PRICE
-------------------------------------------------- ---------- ----------
  SUBTOTAL
----------


10 rows selected.
```

Activ